

NVision: A Sequential Monte Carlo Framework for Bayesian Experiment Design in Nitrogen-Vacancy Center Magnetometry

NVision Project¹

¹*Simulation framework described herein; physical implementation not performed.*

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Optically detected magnetic resonance (ODMR) of nitrogen-vacancy (N-V⁻) centers in diamond is a leading platform for nanoscale magnetometry, but conventional frequency-swept measurement wastes most of its budget sampling far from resonance. Sequential Bayesian experiment design (SBED) instead chooses each measurement to maximize expected information gain (EIG) about the resonance parameters [1]. This paper describes *NVision*, which extends this idea with a Sequential Monte Carlo (SMC) particle belief on a dimension-agnostic unit cube, scaling to Zeeman-split, hyperfine-split, and Voigt-broadened lineshapes with up to seven parameters. An EIG-maximizing acquisition rule, a closed-form Cramér–Rao lower bound (CRLB) per lineshape, and a multi-criterion stopping test jointly decide where to measure and when to stop. We present the signal models, the inference/acquisition mathematics, the simulation-harness architecture, and the evaluation metrics, all cross-checked against the reference implementation.

I. INTRODUCTION

Nitrogen-vacancy (N-V⁻) centers are diamond point defects with a spin-1 ground state whose $m_s = \pm 1$ sublevels Zeeman-split under an external field (Fig. 1); spin-dependent fluorescence makes this splitting optically readable as dips in an ODMR spectrum [2–4]. This makes N-V⁻ magnetometry a widely used nanoscale and vector field-sensing tool [5].

Conventional ODMR sweeps frequency over a fixed grid, wasting most measurements far from resonance where they barely change the fit uncertainty. Dushenko *et al.* [1] showed that choosing each frequency to maximize a posterior-based utility function instead — SBED — cuts the required measurements by more than an order of magnitude, using a two-parameter (frequency, linewidth) grid posterior and the **OptBayesExpt** library.

NVision generalizes this along three axes absent from that work and from an earlier internal HUI course attempt at the same problem [6] (Table I; bold = new here).

The particle-based unit-cube belief (Section III) is what makes the richer, multi-parameter lineshapes of Section II tractable in place of the grid both prior attempts share; the CRLB-derived stopping rule (Sections V–VI) is the paper’s most concrete methodological contribution, replacing a hand-tuned tolerance with one grounded in the measurement’s own information content. The EIG-maximization principle itself is Dushenko *et al.*’s [1, 7], and unlike that work — validated on real N-V⁻ hardware — NVision is simulation-only (Section VIII).

Section II covers the physical models and simulation architecture; Sections III–VI the inference/acquisition math; Section VII the evaluation metrics; Section VIII scope and limitations.

TABLE I. What NVision introduces relative to the two prior attempts. Bold = new here; plain = inherited.

Aspect	Prior attempts	This work
Posterior	OptBayesExpt grid, ≤ 2 –3 params	<i>N</i>-particle SMC, unit cube, up to 7
Signal models	Bare Lorentzian / one triplet	4 lineshape fami- lies (hyperfine, Zeeman, Voigt, saturation)
Stopping rule	Fixed tolerance / iteration count	Closed-form per- lineshape CRLB budget

II. PHYSICAL MODEL AND EXPERIMENT LAYOUT

A. N-V center structure and Zeeman splitting

An external field B along the N-V axis splits $m_s = \pm 1$ symmetrically about the zero-field frequency $f_B \approx 2.87$ GHz by $\gamma_{\text{NV}}B$ ($\gamma_{\text{NV}} \approx 28$ GHz/T), producing two Zeeman dips (Fig. 1); each is further split into a ¹⁴N hyperfine triplet (≈ 2.16 MHz) with finite homogeneous and inhomogeneous linewidth. Section II B’s lineshapes implement this hierarchy as composable, independently toggleable variants.

B. Signal models

All models are built around a single-Lorentzian-dip kernel

$$L(x; f, \Omega) = \frac{\Omega^2}{\Omega^2 + (x - f)^2}, \quad (1)$$

unit height at $x = f$, HWHM Ω (FWHM = 2Ω).

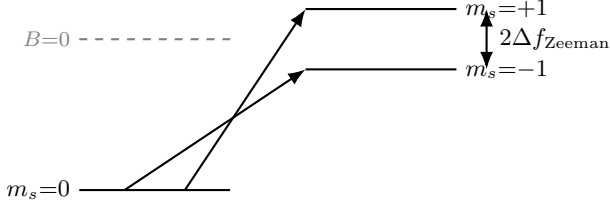


FIG. 1. NV^- ground-state spin triplet: field B splits $m_s = \pm 1$ symmetrically about $f_B \approx 2.87$ GHz by $\gamma_{\text{NV}}B$; microwave transitions from $m_s=0$ (arrows) give the two Zeeman-split ODMR dips.

$$p_{\text{sum}} = \frac{1}{k_{\text{np}}} + 1 + k_{\text{np}}, \quad p_0 = \frac{c}{p_{\text{sum}}}, \quad p_L = \frac{c}{k_{\text{np}} p_{\text{sum}}}, \quad p_R = \frac{c k_{\text{np}}}{p_{\text{sum}}}, \quad (p_L + p_0 + p_R \equiv c), \quad (2)$$

$$S(x) = 1 - \left[p_L L(x; f_0 - \Delta f, \Omega) + p_0 L(x; f_0, \Omega) + p_R L(x; f_0 + \Delta f, \Omega) \right], \quad (3)$$

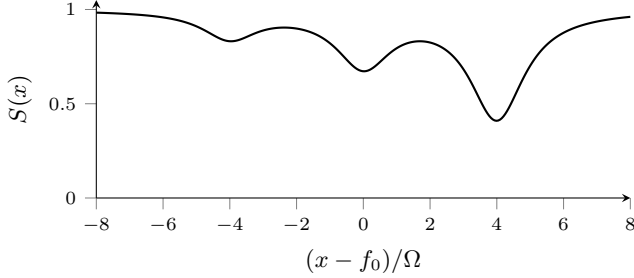


FIG. 2. Lorentzian triplet, Eq. (3), for $k_{\text{np}} = 2$, $\Delta f/\Omega = 4$: weights $p_L=c/7, p_0=2c/7, p_R=4c/7$ ($c=1$) sum exactly to c .

$k_{\text{np}} \in [1, 5]$. Unlike Eq. (1) of Dushenko *et al.* [1], where k_{np} scales two raw amplitudes without renormalizing, k_{np} here only redistributes c between dips. With hyperfine splitting off, Δf is fixed to 2.16 MHz and $k_{\text{np}} = 1$ (equal-weight triplet): the code calls this the “single dip” configuration, but Eq. (3) is still evaluated at nonzero splitting — only the number of *inferred* parameters drops (3 vs. 5 with hyperfine on), not the underlying triplet structure, which visually blends into one dip only once $\Omega \gtrsim \Delta f$. With Zeeman splitting on, each dip weight in Eq. (2) is halved and the triplet duplicated at $f_0 \mp \Delta f_{\text{Zeeman}}$ (up

a. *Lorentzian triplet* (`NVCenterLorentzianModel`). With hyperfine splitting on, `nv_center_lorentzian_eval` uses a population-normalized reparameterization so the three dip weights always sum to the total contrast c (Fig. 2):

to six free parameters, two dip clusters).

b. *Pseudo-Voigt triplet* (`NVCenterVoigtModel`). Each dip is a two-width pseudo-Voigt profile (`nv_center_pseudo_voigt_eval`), height-normalized as

$$V(x) = \frac{e_{\text{lf}}}{x^2 + \gamma_{\text{hom}}^2} + e_{\text{gf}} \exp\left(-\frac{x^2}{2\sigma_{\text{inhom}}^2}\right), \quad (4)$$

with $\gamma_{\text{hom}} = \frac{1}{2}\ell_{\text{frac}}\Gamma$, $\sigma_{\text{inhom}} = \frac{1}{2}(1-\ell_{\text{frac}})\Gamma/\sqrt{2\ln 2}$ splitting total FWHM Γ by Lorentzian fraction $\ell_{\text{frac}} \in [0, 1]$, and $e_{\text{lf}}, e_{\text{gf}}$ fixed via the Thompson–Cox–Hastings weight η so $V(0) = 1$ — algebraically the code’s on-the-fly normalization, pre-factored. This is not a true Voigt profile (no Faddeeva function), and η is calibrated for a shared FWHM rather than the split widths used here, so the true-Voigt approximation error is uncontrolled.

Like the Lorentzian family, this triplet is population-normalized, reusing the same (p_L, p_0, p_R) weights of Eq. (2) (unified onto this convention 2026-07-15 — an earlier per-dip `dip_depth` convention, not population-normalized, is retained only for the single, non-degenerate zero-field peak, `NVCenterOnePeakLorentzianSpectrum`):

$$S(x) = 1 - \left[p_L V(x - f_0 + \Delta f) + p_0 V(x - f_0) + p_R V(x - f_0 - \Delta f) \right], \quad (5)$$

so total contrast c stays fixed as k_{np} varies, exactly as in Eq. (3). The saturation-coupled model below reuses this same population kernel ($\Delta f_{\text{Zeeman}}=0$ when Zeeman

splitting is off).

c. *Saturation-coupled* (`NVCenterSaturationVoigtModel`). Linewidth and *Voigt*

contrast are coupled through one drive parameter s (power relative to half-saturation), since both are set by the same microwave field:

$$\gamma_{\text{hom}}(s) = \gamma_0 \sqrt{1+s}, \quad C(s) = c_{\text{max}} \frac{s}{1+s}, \quad (6)$$

with fixed zero-power HWHM $\gamma_0 = 150$ kHz and saturated contrast c_{max} . The independent Gaussian width σ_{inhom} is uncoupled to drive power. $(s, \sigma_{\text{inhom}}, c_{\text{max}})$ reparameterizes into $(\Gamma, \ell_{\text{frac}}, c)$ of Eq. (4) before evaluation; $\sigma_{\text{inhom}} \rightarrow 0$ recovers a pure Lorentzian exactly.

C. Physical parameter ranges

Table II lists the physical bounds shared by the signal generator and the SMC prior, so the two cannot drift apart.

D. Simulation harness architecture

Each simulated experiment is a Generator×Noise×Locator task (Fig. 3): a ground-truth signal (Section II B), an additive Gaussian noise floor (this paper’s scope; Poisson and Rao-Blackwellized inverse-gamma models exist but are not covered), and an acquisition-and-stopping strategy (uniform/Sobol baselines or a Bayesian locator). The Runner executes many tasks concurrently over independent, seed-deterministic, resumable repeats, streaming results to an on-disk cache for large repeat counts, and automatically runs a matched Sobol baseline for every Bayesian strategy under identical noise and ground truth. Within a repeat, `next()/observe()/done()` iterate until the stopping criteria of Section VI fire, recording the metrics of Section VII.

III. SEQUENTIAL MONTE CARLO BELIEF

The posterior over d free parameters θ is N weighted particles $\{(\theta_i, w_i)\}_{i=1}^N$, scaling to $d > 2$ without the grid’s

$$\tilde{w}_i \leftarrow w_i p(y \mid \theta_i, x), \quad w_i \leftarrow \tilde{w}_i / \sum_j \tilde{w}_j,$$

where S is the signal at particle i ; the σ -only normalization constant cancels under renormalization. A tempering factor β rescales $\sigma \rightarrow \sigma/\sqrt{\beta}$; the update runs in

exponential memory blow-up — the mechanism enabling Section II B’s six-parameter models.

A. Unit-cube / physical coordinate separation

All particle arithmetic happens on a dimension-agnostic unit cube $[0, 1]^d$; each parameter j maps to its physical bound $[l_j, h_j]$ via a fixed affine rescale map,

$$\theta_j^{\text{phys}} = l_j + u_j (h_j - l_j), \quad \sigma_j^{\text{phys}} = \sigma_j^u (h_j - l_j), \quad (7)$$

owned exclusively by the belief and fixed for a run’s lifetime. This is kept strictly separate from the *focus window* (Section IV E) — the physical sub-interval the locator currently probes, which narrows over a run but never

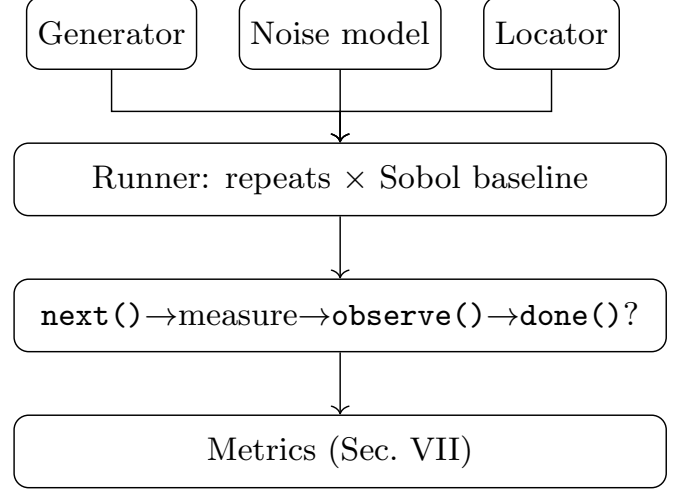


FIG. 3. Simulation harness: a Generator×Noise×Locator task runs under the Runner across independent repeats (Sec. II D), iterating until the stopping criteria of Sec. VI fire.

touches Eq. (7): the belief only sees the rescale map, acquisition/observation code only sees the focus window, preventing a class of coordinate-transform bugs.

B. Weight update

For observation (x, y) under Gaussian noise σ , each particle’s weight updates multiplicatively,

$$\log p(y \mid \theta_i, x) = -\frac{1}{2} \left(\frac{y - S(x, \theta_i)}{\sigma} \right)^2, \quad (8)$$

log-space with a max-shift to avoid underflow.

TABLE II. Representative physical parameter bounds (Lorentzian/Voigt family).

Parameter	Symbol	Range
Center frequency domain	f_0	2.72–3.02 GHz
Linewidth (HWHM)	Ω	200 kHz–5.0 MHz
Hyperfine splitting	Δf	2.0–8.5 MHz
Nuclear-polarization asymmetry	k_{np}	1.0–5.0
Zeeman half-separation	Δf_{Zeeman}	0–60 MHz ($\approx$ 0–2.1 mT)
Natural (zero-power) HWHM	γ_0	150 kHz (fixed)
^{14}N hyperfine constant	—	2.16 MHz (fixed, when not inferred)
Saturation parameter	s	0.02–30

C. Resampling

ESS = $1/\sum_i w_i^2$ triggers systematic resampling when ESS < $r_{\text{thr}}N$ ($r_{\text{thr}} = 0.20$): a single $U \sim \text{Uniform}(0, 1/N)$ gives positions $u_k = (k+U)/N$, mapped to particle indices by walking the weight CDF — lower variance than multinomial resampling.

D. Liu–West kernel

Post-resample particles contract toward the weighted mean $\boldsymbol{\mu}$ and jitter (Liu–West [8], with an exploration floor and rejuvenation):

$$\boldsymbol{\theta}_i \leftarrow a\boldsymbol{\theta}_i + (1-a)\boldsymbol{\mu} \quad (\text{shrinkage}), \quad (9)$$

$$\boldsymbol{\theta}_i \leftarrow \boldsymbol{\theta}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, (1-a^2)\boldsymbol{\Sigma}) \quad (\text{nudge}), \quad (10)$$

with $a = 0.98$, $\boldsymbol{\Sigma}$ the pre-resample covariance. The nudge covariance diagonal is floored,

$$C_{jj} \leftarrow \max\left(C_{jj}, [(h_j - l_j) f_{\text{expl}} e^{-t/25}]^2\right), \quad f_{\text{expl}} = 0.01, \quad (11)$$

so exploration decays but never vanishes (t = step count); eigen-regularization (min eigenvalue $\max(10^{-11}, 10^{-6}\lambda_{\text{max}})$) then guarantees a valid Cholesky

factor, and a decaying fraction $f_{\text{rejuv}} = 0.05 e^{-t/25}$ of particles are replaced by fresh prior draws to prevent mode collapse.

E. Weighted posterior statistics

Estimates and uncertainties are the weighted mean/std,

$$\mu_j = \frac{\sum_i w_i \theta_{ij}}{\sum_i w_i}, \quad \sigma_j = \sqrt{\frac{\sum_i w_i (\theta_{ij} - \mu_j)^2}{\sum_i w_i}}, \quad (12)$$

and the weighted covariance $\boldsymbol{\Sigma}$ feeds both the Liu–West kernel and the Gaussian entropy approximation

$$H \approx \frac{1}{2} \ln |\boldsymbol{\Sigma}| + \frac{d}{2} (1 + \ln 2\pi), \quad (13)$$

via a ridge-stabilized `slogdet`.

IV. SEQUENTIAL BAYESIAN EXPERIMENT DESIGN

A. Expected information gain

The acquisition rule approximates the EIG of candidate x by the particle cloud’s predictive variance,

$$\text{EIG}(x) \approx \frac{1}{2} \ln \left(1 + \frac{\sigma_{\text{pred}}^2(x)}{\sigma_{\text{noise}}^2} \right), \quad \sigma_{\text{pred}}^2(x) = \sum_i w_i S(x, \boldsymbol{\theta}_i)^2 - \left(\sum_i w_i S(x, \boldsymbol{\theta}_i) \right)^2, \quad (14)$$

with $\sigma_{\text{noise}}^2 = \max(\sigma^2, 10^{-12})$ — the exact entropy reduction for a Gaussian observation $y = S + \eta$, the same utility motivating SBED generally [1, 7]. Eq. (14) is evaluated on a stratified subsample of $\leq N_{\text{EIG}} = 500$ particles; between resamples the prediction matrix $M_{ci} = S(x_c, \boldsymbol{\theta}_i)$ is cached, so only $\sigma_{\text{pred}}^2 = M^{(2)}\mathbf{w} - (M\mathbf{w})^2$ is recomputed as weights change.

Candidates come from a slope-targeted epoch grid (dense windows at each dip’s $\pm$ HWHM points, plus a coarse global grid) rebuilt each resample. EIG scores are chunked (64 candidates); each chunk’s arg-max competes via a softmax at $\tau = 0.01$, avoiding lock-on to a single numerical-noise spike.

B. Exploration / dip-bias / EIG mixture

Each step, one uniform draw selects among three branches (Table III), so early search is not fully dependent on a possibly-biased posterior.

C. Background noise estimation

The noise std is not assumed known; it is estimated online from *background* measurements ($|x - \hat{f}| > k_{\text{bg}}|\hat{\Omega}|$, $k_{\text{bg}} = 3.0$) via the median absolute deviation,

$$\hat{\sigma}_{\text{bg}} = 1.4826 \text{ median}(|y_i^{\text{bg}} - \text{median}(y^{\text{bg}})|), \quad (15)$$

$$n_{\text{theory}} = \frac{4\hat{\sigma}_{\text{bg}}^2 \hat{\Omega} W}{\pi \hat{c}^2 T^2}, \quad N_{\text{budget}} = \max(N_{\text{max}}, K_{\text{theory}} n_{\text{theory}} + 1), \quad (16)$$

with $T = 100$ kHz, W the bandwidth, $K_{\text{theory}} = 20$ — large enough that this fires only when EIG acquisition has failed to converge far sooner.

E. Focus-window narrowing

Each resample narrows the search window to the union of particles' predicted active regions, $[f_i - \Delta f_i - k\Omega_i, f_i + \Delta f_i + k\Omega_i]$ ($k = 3.0$), using the 1st/99th-percentile edges to resist outliers; if particles pile against a unit-cube boundary ($>15\%$ within 5%), the window expands instead. This never mutates the fixed rescale map (Eq. 7).

V. FISHER INFORMATION AND THE CRAMÉR–RAO LOWER BOUND

A. General Gaussian Fisher information

For observation x under Gaussian noise, $\mathbf{I}(\boldsymbol{\theta}; x) = \sigma^{-2} \nabla_{\boldsymbol{\theta}} S \nabla_{\boldsymbol{\theta}} S^{\top}$; cumulative $\mathbf{I}_{\text{cum}} = \sum_{n=1}^N \mathbf{I}(\boldsymbol{\theta}; x_n)$, and $\text{CRLB}_j = \sqrt{[(\mathbf{I}_{\text{cum}} + \epsilon \mathbf{I})^+]_{jj}}$ ($\epsilon = 10^{-6}$) via a ridge-regularized Moore–Penrose pseudo-inverse [9].

B. Closed-form CRLB for a uniformly sampled Lorentzian

For $S = cL(x; f, \Omega)$ sampled at uniform density $\rho = N/W$,

$$\int_{-\infty}^{\infty} \left(\frac{\partial L}{\partial f} \right)^2 dx = \frac{\pi}{4\Omega}, \quad (17)$$

with $1.4826 = 1/\Phi^{-1}(3/4)$ the consistent-estimator factor. Below 15 background points, forced calibration samples the two background regions (weighted by width) until Eq. (15) is estimable.

D. Theoretical step-budget backstop

Once $\hat{\sigma}_{\text{bg}}$, $\hat{c} > 0$ are available, a permissive backstop (Section VB) bounds runaway,

so $I(f) = \rho c^2 \pi / (4\sigma^2 \Omega)$ and

$$\boxed{\text{Var}^{\text{CRLB}}(f) = \frac{4\sigma^2 \Omega}{\pi c^2 \rho}}, \quad \text{CRLB}_f = \sqrt{\text{Var}^{\text{CRLB}}(f)}, \quad (18)$$

uncertainty growing with σ, Ω and shrinking with c, ρ , as expected.

The same approach on the pseudo-Voigt (Eq. 4), dropping the Lorentzian–Gaussian cross-term (conservative: overestimates Var^{CRLB} for mixed lineshapes), gives

$$J = e_{\text{lf}}^2 \frac{\pi}{4\gamma_{\text{hom}}^5} + e_{\text{gf}}^2 \frac{\sqrt{\pi}}{2\sigma_{\text{inhom}}}, \quad \text{Var}^{\text{CRLB}}(f) = \frac{\sigma^2}{\rho c_{\text{total}}^2 J}, \quad (19)$$

reducing exactly to Eq. (18) as $\sigma_{\text{inhom}} \rightarrow 0$; both forms were checked against direct numerical integration of the coded kernels.

C. Numerical gradients revive the per-parameter FIM

None of the N-V[−] signal models expose an analytical $\nabla_{\boldsymbol{\theta}} S$, so CRLB_j (Eq. (18)'s general form, Section V) returned $\emptyset$ for every model in practice; the multi-parameter FIM machinery was present in the code but never exercised. A central-difference fallback, with step size scaled by each parameter's own bound range rather than its value (parameters here span ~ 7 orders of magnitude, from Hz-scale widths to a dimensionless contrast ≈ 0.25) and clamped into bounds, makes $\mathbf{I}_{\text{cum}}$ non-trivial for these models. Since the SMC belief and the FIM must share one coordinate system, the gradient point is taken in whatever coordinates the belief's model actually uses (unit-cube for `UnitCubeSMCMarginalDistribution`), with CRLB_j

TABLE III. Acquisition branch selection at step t .

Draw u	Action	Purpose
$u < 0.1 e^{-t/25}$	Uniform global sample	Decaying exploration
$u < 0.20$	Near empirical dip ± 5 MHz	Correct posterior bias
otherwise	Maximize EIG (Eq. 14)	Main information path

rescaled back to physical units by each parameter’s range afterward.

With the FIM live, non-frequency parameters in `reported_uncertainty()` are floored at their own marginal CRLB unconditionally (independent of the stopping-rule question addressed next): $\sigma_j^{\text{reported}} = \max(\sigma_j^{\text{phys}}, \text{CRLB}_j)$. Measured over 120 mixed repeats (56 deliberately degenerate), this moves median error/ σ^{reported} for `zeeman_split` from 1.21 to 0.90 and the fraction beyond 3σ from 8.3% to 2.5% (`c_total`: 1.61 $\rightarrow$ 0.96, 18.3% $\rightarrow$ 3.3%) — the particle spread alone understates uncertainty exactly on the near-flat ridge where `zeeman_split` trades off against the width pair, while the FIM correctly reports a large CRLB there because it is near-singular in that direction. No safety factor is applied, since the CRLB is already a hard lower bound; a $4\times$ factor (matching K_{safety}) was measured to

over-correct badly (medians drop to 0.09–0.25).

VI. CONVERGENCE CRITERIA

A run stops when *all* of the following hold for 8 consecutive resamples (`convergence_patience_steps`; any failure resets the streak, so a transient false positive cannot end a run early):

a. Per-parameter thresholds. Frequency: absolute $\sigma_f < T_f = 100$ kHz; others: relative 1% of bound width unless overridden. For saturation-Voigt, raw ($s, \sigma_{\text{inhom}}, c_{\text{max}}$) are not gated directly (inconsistent along the nonlinear saturation axis); instead their uncertainty propagates via the local Jacobian onto effective HWHM $\Omega(s, \sigma_{\text{inhom}}) = \gamma_0 \sqrt{1 + s} + \sqrt{2 \ln 2} \sigma_{\text{inhom}}$ and realized contrast $C(s, c_{\text{max}}) = c_{\text{max}} s / (1 + s)$, gated with ordinary linewidth/contrast semantics.

b. Overall RMS check. Even if every parameter passes individually,

$$u_j = \sigma_j / B_j, \quad B_j = \begin{cases} T_f / \text{threshold} & j = \text{frequency} \\ h_j - l_j & \text{otherwise} \end{cases}, \quad \text{RMS} = \sqrt{\frac{1}{d} \sum_j u_j^2} < \text{threshold}, \quad (20)$$

must also pass, so one marginal parameter cannot mask the rest.

c. Dynamic CRLB-derived stopping. Since $\text{CRLB}_f \propto 1/\sqrt{t}$, steps still needed are $n_{\text{req}} = t(\text{CRLB}_f / T_f)^2$; budget $N_{\text{dyn}} = \min(N_{\text{max}}, K_{\text{safety}} n_{\text{req}} + 1)$ ($K_{\text{safety}} = 4.0$), stopping as infeasible if $n_{\text{req}} > N_{\text{max}} K_{\text{safety}}$. This closed-form, frequency-only rule is unconditionally active. In SBED, the same idea extends to every parameter using the numerical-gradient FIM of Section VC, rescaled by the online noise estimate, $\text{CRLB}_j^{\text{scaled}} = \text{CRLB}_j^{\text{FIM}} \hat{\sigma}_{\text{bg}} / \sigma_{\text{nominal}}$, stopping when $\sigma_j < K_{\text{safety}} \text{CRLB}_j^{\text{scaled}}$ for every target parameter — but, as the next paragraph reports, this multi-parameter gate is kept *off* by default.

d. Why the multi-parameter FIM gate is disabled by default. Turning it on once Section VC’s fallback made CRLB_j non-trivial measurably improves the benchmark numbers — median run length 450 $\rightarrow$ 24 steps, catastrophic-error rate on known-degenerate configurations 67% $\rightarrow$ 12% — but we found this to be an artifact of the evaluation harness rather than a genuine capa-

bility. Near a degenerate point, the marginal CRLB is inflated by a near-singular $\mathbf{I}_{\text{cum}}$ (11.8 MHz measured on `zeeman_split` where the achieved error was only 1.3 MHz), so the stopping test passes trivially from the first few steps — exactly where the problem is hardest. The apparent accuracy gain traces to the generators’ own prior: each repeat’s SMC prior mean is drawn as $\mathcal{N}(\theta_{\text{true}}, \sigma_{\text{prior}})$, so stopping early scores well because the run reports a truth-centred prior it never had to earn against real measurements. This is the same failure class as three earlier snapshot-CRLB early-stop bugs found during development of this locator, and we flag it here as a caution against validating simulation-based stopping rules purely against simulated-prior-centred accuracy.

e. Adaptive plateau stopping (default). In place of the FIM gate, the shipped default stops a run once every target parameter’s *point estimate* has stopped moving relative to its own current uncertainty — measuring diminishing returns directly rather than trusting a derived quantity’s claim about the information limit. Writing

$\hat{\theta}_j^{(t)}$ for the estimate at check t ,

$$\text{plateaued}_j \iff |\hat{\theta}_j^{(t)} - \hat{\theta}_j^{(t-W)}| < f_\sigma \sigma_j^{(t)}, \quad (21)$$

with window $W = 30$ checks and fraction $f_\sigma = 0.25$; a run stops once every target parameter plateaus for `convergence_patience_steps` consecutive checks. Movement is scaled by each parameter’s own σ (not its bound range), so one threshold applies across parameters spanning orders of magnitude. On the N-V Voigt grid (120 repeats), median `zeeman_split` error falls 1.483 MHz (step 10) $\rightarrow$ 0.237 (25) $\rightarrow$ 0.059 (50) $\rightarrow$ 0.022 (100) $\rightarrow$ 0.016 (450) MHz — a knee near step 100, after which the remaining 350 of a 450-step budget buy $\sim 1.3\times$ accuracy for $4.5\times$ the measurements. $f_\sigma = 0.25$ was calibrated by replaying the rule inside full-budget runs, firing on 100% of both ordinary and degenerate configurations and saving $2.3\times$ and $3.6\times$ measurements respectively (median step 450 $\rightarrow$ 197 ordinary, 450 $\rightarrow$ 124 degenerate). A separate live A/B (120 configs/arm) confirmed the rule end-to-end rather than by replay, giving a close but distinct 450 $\rightarrow$ 220 ordinary (124 degenerate, unchanged); accuracy on most target parameters stayed within a few percent of the full-budget run — well under the $\sqrt{2}$ – $\sqrt{3.6}$ increase a naive $\sqrt{n}$ scaling would predict. The one measured cost, from that same live A/B: on degenerate configurations the catastrophic-error rate rises 7.1% $\rightarrow$ 12.5% (4 $\rightarrow$ 7 of 56 repeats); a more conservative $f_\sigma = 0.15$ keeps most of that margin at a $2.2\times$ rather than $3.6\times$ saving.

Table IV summarizes the defaults above.

Novelty. The Fisher/Cramér–Rao construction is textbook [9]; new here are (i) closed-form CRLBs for this lineshape family (Eqs. 18–19), absent from either prior attempt or Dushenko *et al.* [1], used as the frequency-only stopping/feasibility mechanism in place of fixed-iteration or fixed-tolerance rules — arguably the paper’s most concrete methodological contribution — and (ii) the negative result that the natural multi-parameter extension of the same idea (gate every parameter on its own FIM-derived CRLB) is not safe to use as-is under a benchmark with an informative simulated prior, together with the adaptive plateau rule we use instead, which achieves a comparable 2–3.6 $\times$ measurement reduction without that failure mode.

VII. EVALUATION METRICS

A. Point-estimate error

For a single peak, $\text{abs_err_x} = |\hat{f} - f_{\text{true}}|$; for two peaks (e.g. a resolved Zeeman doublet), estimates and truths are sorted and paired before computing $\text{pair_rmse} = \sqrt{\frac{1}{2}(\text{abs_err_x}_1^2 + \text{abs_err_x}_2^2)}$.

B. Uniform-sweep baseline

To quantify savings, each run is compared against the uniform measurement count needed for the *same* ground truth. Nyquist argument: a feature of width w is resolvable at spacing $\delta \leq w/2$, giving

$$n_{\text{sweep}} = \frac{2W}{w_{\text{eff}}}, \quad (22)$$

with w_{eff} measured on the true signal (20,000-point grid, 20%-depth threshold, segments merged within 2% of the domain) rather than a worst-case bound. E.g. $W = 500$ MHz, $\Omega \approx 100$ kHz ($w = 4\Omega \approx 400$ kHz) gives $n_{\text{sweep}} \approx 2500$ — scaling as W/Ω , not fixed. Reported $\text{sobol_difference} = n_{\text{sweep}} - n_{\text{measured}} > 0$ whenever SBED used fewer measurements.

C. Failure classification

Each run gets one failure reason, in priority order: success (frequency converged, or an unconverging baseline); `infeasible_crlb` (pre-run gate tripped); `timeout`; `theory_budget` (Section IV count exceeded); `max_steps` (budget exhausted otherwise). The `infeasible_crlb` gate reuses `marginal_crlbs_at_budget`, which — like `crlb_per_param()` before Section VC’s fallback — requires an analytical gradient and was not extended to the numerical fallback; for every N-V[−] model in this paper’s scope it therefore returns no CRLBs and this gate never actually fires, so `infeasible_crlb` does not occur among the results reported here.

VIII. DISCUSSION

NVision keeps Dushenko *et al.*’s central claim — EIG maximization converges in far fewer measurements than a fixed sweep — but swaps the two-parameter grid for an N -particle SMC belief, trading grid intractability for particle noise at small N (mitigated by Section III’s resampling/shrinkage machinery and the CRLB backstop of Sections V–VI).

Two approximations bound these claims: the pseudo-Voigt (Eq. 4) is not a true Voigt profile, with uncontrolled deviation from one; and the Voigt CRLB (Eq. 19) drops the Lorentzian–Gaussian cross-term, conservative but inexact for strongly mixed lineshapes. Both are checked numerically against the coded kernels, not an independent analytic Voigt CRLB. This paper’s scope is the additive Gaussian noise path; Poisson and Rao–Blackwellized inverse-gamma noise models exist in the code but are not covered. Finally, metrics are computed against the true signal’s resolvable width rather than a worst-case parameter bound, since the same Ω range (200 kHz–5 MHz) spans an order of magnitude in sweep budget.

A further caveat concerns any stopping-rule accuracy claim measured on this harness: because each repeat’s

TABLE IV. Key inference-stack constants and their defaults.

Symbol	Meaning	Default
N	SMC particle count	1000
r_{thr}	ESS resampling threshold (fraction of N)	0.20
a	Liu–West shrinkage	0.98
N_{EIG}	EIG subsample size	500
T_f	Frequency convergence threshold	100 kHz
K_{safety}	CRLB safety factor	4.0
K_{theory}	Theory-budget safety factor	20
k_{bg}	Background classification span (linewidths)	3.0
patience	Consecutive converged resamples required	8
threshold	Relative convergence threshold	1%
W	Plateau-stop window (checks)	30
f_{σ}	Plateau-stop drift fraction	0.25

SMC prior is drawn centred on that repeat’s own ground truth (Section III), a rule that stops as soon as some derived quantity looks converged can score well by reporting a prior it never had to earn, not by having actually resolved the signal — Section VI’s FIM-gate finding is one concrete instance of this, and it should be treated as a general risk when validating any stopping criterion, CRLB- or otherwise, purely against simulated-prior-centred error.

lineshapes via a dimension-agnostic SMC belief. An EIG acquisition rule, online noise estimation, and per-lineshape closed-form CRLBs jointly decide where to measure and when to stop; the accompanying harness runs this at scale against Nyquist-derived uniform-sweep and Sobol baselines on matched ground truth, with derivations cross-checked against the reference implementation throughout.

IX. CONCLUSION

NVision generalizes single-dip SBED for N-V[−] ODMR [1] to hyperfine, Zeeman, Voigt, and saturation-coupled

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